Claire M. Williams

Department of Biology at Georgia Tech and
Department of Ecology and Conservation Biology at Texas A&M
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Research Interests: My research explores how marine species and communities respond to environmental and anthropogenic change across ecological and geological timescales. I combine field ecology, water-quality monitoring, trait-based approaches, species distribution modeling, and paleoecology to examine how environmental change restructures marine communities and their functions to place modern reef conditions within longer-term ecological and climatic contexts. My research aims to establish realistic conservation baselines and improve predictions of how marine ecosystems will respond to future environmental change.

EDUCATION AND PROFESSIONAL APPOINTMENTS/EMPLOYMENT

2026 Postdoctoral Researcher
Department of Ecology and Conservation Biology at Texas A&M and
School of Biological Sciences at Georgia Tech

2025 PhD in Geological Sciences,
University of Texas at Austin
Thesis: Environmental Controls and Impacts on Caribbean Reefs from the Last Interglacial to the Future

Advanced Teaching Preparation Certificate

2020 B.S. in Integrative Biology Honors, High Distinction, *Summa Cum Laude*
University of Illinois Urbana-Champaign
Thesis: The impact of the invasive ant *Tetramorium immigrans* on myrmecochory in Illinois

B.S. in Geology, Highest Distinction
University of Illinois Urbana-Champaign
Thesis: Morphological changes in *Cyclocardia granulata* following Plio-Pleistocene extinction in Virginia and North Carolina, Western Atlantic

Minor in Chemistry

Certificate in Undergraduate Research

PUBLICATIONS

Undergraduate Student[UG]

Published

2026 McGuire, J. L., **Williams, C. M.,** When mammals crossed between continents. *Science Perspective,* 393,458-459. https://doi.org/10.1126/science.aei5362

2026 **Williams, C. M.,** Rahamut, M., Martindale, R. C. Pleistocene reef composition documents a healthy ecological baseline for a sanctuary in northeast Jamaica. *PALAIOS*. 41(4), 215–226. http://dx.doi.org/10.2110/palo.2025.041

2026 **Williams, C. M.,** Machado-Stredel, F., Martindale, R. C., Myers, C., Integrating fossil data in ecological niche models improves predictions of future habitat demonstrated with Caribbean reef corals. *Conservation Biology,* e70323. http://doi.org/10.1111/cobi.70323

2025 **Williams, C. M.,** Henry, D., Martindale, R. C., & Gordon-Smith, D.-A. Spatiotemporal variability in reef ecology and herbivore impacts in the East Portland Special Fishery Conservation Area, northeast Jamaica (2017 – 2024). *Frontiers in Marine Science*, 12. https://doi.org/10.3389/fmars.2025.1684741

2024 Sulbaran Reyes, B. S.[UG], Martindale, R. C., Salgado-Jauregui, E., Sinha, S., **Williams, C. M.,** & Cooc, N. Factors that promote cooperation, learning, and engagement with geoscience concepts when playing educational board games in an entry-level college paleontology lab. *Journal of Geoscience Education*, 1–19. https://doi.org/10.1080/10899995.2024.2401749

Manuscripts in Preparation

In Preparation **Williams, C. M.,** McGuire, J. L., Bolden, I. W., Lawing, A. M. An Ecometric Framework for Anticipating Reef Collapse and Recovery. Manuscript in Prep to be submitted November 30, 2026. *Proceedings of the Royal Society B*

In Preparation **Williams, C. M.,** Gordon-Smith, Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Environmental monitoring to support reef sanctuary management in East Portland Jamaica. Manuscript in Prep to be submitted December 20, 2026. *Marine Pollution Bulletin*

In Preparation **Williams, C. M.,** Tyler, C., Martindale, R. C., and Yellman, Z.[UG] Disease and bleaching drive functional erosion from historical reef baselines in northeast Jamaica. Manuscript in Prep to be submitted December 30, 2026. *Nature communications Earth & Environment*

Other Publications and Creative Products

2026 Curriculum Development
Martindale, R. C., **Williams, C. M.,** Currie, C.[UG], Kemp, M. "Reef Survivor Jr" - Board Game about reef ecosystems for Grades 4-6 https://serc.carleton.edu/teachearth/activities/312725.html

2025 Curriculum Development
Williams, C. M., & Rempel, H. Coral Reefs & Our Changing Oceans: Three-part Grades 4–5 science curriculum aligned with Texas Essential Knowledge and Skills (TEKS). Open Educational Resources Commons. https://oercommons.org/courseware/lesson/135980/overview

Other publications in Review

In Review Martindale R. C., Salgado-Jauregui E., **Williams, C. M.,** Currie, C.[UG], Sinha, S., Sulbaran Reyes, B. S.[UG], Gordon-Smith, D.-A., Henry, D., Ellins, K., Cooc, N., Coral Reef Board Games Increase Scientific Understanding and Conservation Awareness. *Reef Encounter*

FELLOWSHIPS, GRANTS, AWARDS, AND HONORS ($224,539)

External

2026 Contributed to an accepted NSFGEO-NERC: Identifying the taxon-specific drivers of range dynamics using the quaternary fossil record
2022-2025 NSF Graduate Research Fellowship Program ($159,000)
2023 Conservation Paleobiology Network Student Travel Grant ($1,340)
2022 International Coral Reef Society Ruth Gates Fellowship Honorable Mention
2022 Lerner-Gray Memorial Fund of the American Museum of Natural History ($1,000)
2022 Geological Society of American Graduate Student Research Grant ($2,000)
2022 Paleontological Society Student Research Grant ($1,200)
2022 Society for the Study of Evolution Small Grants Program for Local and Regional Outreach Promoting the Understanding of Evolutionary Biology ($1,000)

Jackson School

2024 Jackson Fellowship in Geohydrology ($8,445)
2024 BIES Summer Research Travel Fellowship ($5,000)
2022-2023 Jackson School of Geoscience Off Campus Research Funds ($6,090)
2023 Jackson School of Geoscience Analytical Fee Fund ($4,464)
2021, 2024 Whitney Endowed Presidential Scholarship ($7,000)
2020-2021 Jackson School of Geosciences Admissions Fellowship ($28,000)

PRESENTATIONS

Presenting author[*] Undergraduate Student[UG]

Invited Presentations

2026 **Williams, C. M.[*],** Improving Baselines to Predict and Recover Coral Reef Function. No Bones Seminar Smithsonian

2025 **Williams, C. M.[*],** Machado-Stredel, F., Martindale, R. C., Myers, C., Integrating Fossil Data in Paleoecological Niche Models Improves Predictions of Future Habitat for Key Caribbean Reef Corals. 4th IFCRS Early Career Researcher Symposium (virtual)

2024 **Williams, C. M.[*],** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Environmental Monitoring to Support Coral Reef Management in the East Portland Special Fishery Conservation Area, Jamaica. Ocean Acidification Week (virtual)

2023 **Williams, C. M.[*],** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in East Portland, Jamaica, University of the West Indies Chemistry Departmental Seminar

Contributed Conference Presentations

2026 Gordon-Smith, D.-A.*, Bolden, I. W., Martindale R. C., **Williams, C. M.,** Bergan, P. The Influence of Submarine groundwater Discharge on Carbonate Chemistry Dynamics of a Coastal Coral Reef System in East Portland Jamaica

2026 **Williams, C. M. *,** McGuire, J. L., Bolden, I. W., Lawing, A. M. An Ecometric Framework for Anticipating Reef Collapse and Recovery. Geological Society of America annual meeting, Denver CO

2025 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in the East Portland Special Fishery Conservation Area. CPEG Meeting & Conservation Palaeobiology Symposium, Zurich

2025 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Tracing Nutrient Stress to Source: Hydrogeologic and Anthropogenic Drivers of Coral Decline in Northeast Jamaica Geological Society of America annual meeting, San Antonio TX

2024 Gordon-Smith, D.-A.*, Bergan, P., **Williams, C. M.,** Martindale R. C. Use Of Radon as a Tracer of Submarine Groundwater Discharge in Coastal Regions of Jamaica. Scientific Research Council Science and Technology Conference, Jamaica

2024 Martindale, R. C.*, Turner, L., Kemp, M., Henry, D., Gordon-Smith, D.-A, Bolden, I. W., **Williams, C. M.**, Bergan, P., Rahamut, M., Haile, B., Chiu, C. Y. [UG], Greene, N. [UG], Hong, K. [UG], Reyes, A. [UG], Salter, G. [UG], Stone, M. [UG], Trench, C., Matheny, A. M., Lessons Learned From The First Year of Geopaths GO Jamaica: Fostering a Learning Ecosystem for Interdisciplinary Earth Scientists Through Service Learning and Conservation in Jamaica. American Geophysical Union Meeting

2024 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in the East Portland Special Fishery Conservation Area, Jamaica. Geological Society of America annual meeting, Anaheim CA

2024 **Williams, C. M.*,** Machado-Stredel, F., Martindale, R. C., Myers, C., Integrating Fossil Data Improves Predictions of Future Habitat for Key Caribbean Reef Corals. North American Paleontological Congress, Ann Arbor MI

2024 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Poster Presentation: Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in the East Portland Special Fishery Conservation Area, Jamaica. National Conference on Ecosystem Restoration, Albuquerque NM

2023 **Williams, C. M.*,** Machado-Stredel, F., Martindale, R. C., Myers, C., Ecological Niche Modeling of Ecologically Important Caribbean Coral Species. Geological Society of America annual meeting, Pittsburgh PA

2023 Currie, C.*,[UG] Dr. Gordon-Smith, D.-A., Ellins, K. Salgado-Jauregui, E., Martindale, R.C. Henry, D., **Williams, C. M.,** Reef Survivor Jamaica- Modification of an Undergraduate Geoscience Board Game for Jamaican Communities (Middle School Children and Adults). Earth Educators' Rendezvous 2023

2023 **Williams, C. M.*,** Machado-Stredel, F., Martindale, R. C., Myers, C., Ecological Niche Modeling of Prominent Caribbean Coral Species. XXI INQUA Congress 2023

2023 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Community Abundance and Environmental Monitoring to Support Coral Reef Management in East Portland Special Fishery Conservation Area, Jamaica. 2nd Conservation Paleobiology Symposium

2022 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Community Abundance Analysis to Support Coral Reef Management in East Portland Special Fishery Conservation Area, Jamaica. International Paleontological Congress, Thailand

2022 **Williams, C. M.*,** Gordon-Smith, D.-A. Bergan, P., Martindale R. C., Henry, D., Community Abundance Analysis to Support Coral Reef Management in East Portland Special Fishery Conservation Area, Jamaica. Geological Society of America annual meeting, Denver CO

2021 **Williams, C. M.*,** Christie, M., Poster Presentation: Morphological Changes in *Cyclocardia granulata* Following the Plio-Pleistocene Extinction in Virginia and North Carolina, Western Atlantic

TEACHING EXPERIENCE

Georgia University of Technology

Guest Lecturer — Fall 2026
- Biodiversity on a Changing Planet
  - Created and taught lecture on corals and spatial methods for studying biodiversity

University of Texas at Austin

Teaching Assistant — Spring 2025
- Life Through Time

Teaching Assistant — Fall 2024
- Earth and Planetary Process Through Time
  - Designed and led a week's lectures and labs on the fossil record and the Pleistocene of Texas

Short Course Instructor for Jackson School of Geosciences Research Traineeship Experience (RTX) — Summer 2021-2022
- Ecological Niche Modeling

STEM camp organizer with Negril Education Environment Trust, University of the West Indies, University of Technology in Jamaica, and the Jamaican Ministry of Education — Summer 2022
- Created and taught lesson plans on Coral Reefs & Our Changing Oceans curriculum and modified board game Reef Survivors to meet Jamaican educational standards for grades 5-8

Teaching Assistant — Fall 2021, Fall 2022
- Python for Geosciences

University of Illinois at Urbana-Champaign

Teaching Assistant — Spring 2019
- General Chemistry II

Teaching Assistant — Fall 2019
- Integrative Biology Honors Organismal Biology

RESEARCH/FIELD EXPERIENCE

2024 — Research Assistant Dr. Rowan Martindale's lab
- Travel to Morocco's High Atlas Mountains to study Jurassic coral reefs. Assisted in fossil coral reef and invertebrate sampling to explore ecosystem recovery after two major reef extinction events.

2019-2021 — Researcher Dr. Max Christie's lab
- Imaged and measured bivalves for 3D morphometrics. Ran morphometric models

2017-2020 — Research Assistant Dr. Andy Suarez's lab
- Measured and imaged the ant genus *Camponotus* skull size and bite strength to determine how bite strength scales with size in polymorphic ants

2019 — Research Assistant Madagascar Ants & Canopy Bootcamp
- Determined the biodiversity of ant species present in the canopy of Ankarafantsika National Park in Madagascar by climbing into the canopy and arboreally sampling trees.

2018 — Research Assistant Research in Bioanthropology
- Recorded how *Cebus capucinus* utilizes disturbed habitat using the activity budget of *Cebus capucinus* and ecologically surveying the disturbance levels of the forest in La Suerte Research Station, Costa Rica

2016-2020 — Illinois Sustainable Technology Center
- Compiled data digitally from research notes from Robertson's study on bees in Carlinville, Illinois to inform policy makers and community members about constructing pollinator plots

SERVICE TO PROFESSION

2026-Present	Organizer of the International Fossil Coral and Reef Society Early Career Researcher Symposium
2023- Present	Webinar Representative for International Conservation Paleobiology Network Early Career Panel
2024	Judge for Capital of Texas Undergraduate Research Conference
2022-2023	Organizer of University of Texas Paleontology Seminar
2021-2022	Jackson School of Geosciences Peer Mentorship Program Mentor
Ongoing	Journal article reviewer Geophysical Research Letters (2022) Royal Society Open Science (2022)

UNDERGRADUATE STUDENTS MENTORED

Present	Isaac Elkins Coral functional traits and trait–environment relationships
2024-Present	Zephyr Yellman Functional restructuring of coral reef communities from the Last Interglacial to the present in a Jamaican marine sanctuary
2024-2025	Angelica Reyes and Kaylin Hong Water Quality along the Rio Grande River, Portland Jamaica
2022-2024	Caitlyn Currie East Portland Water Quality Project Reef Survivor Jamaica- Modification of an undergraduate geoscience board game for Jamaican communities (middle school children and adults
2024-2025	GEOPAths GO! Jamaica, 6 undergraduate mentees Organized and facilitated a multinational field experience for undergraduates, graduate students, researchers, and faculty from the United States, and Jamaica. Created and orchestrated daily field research activities, mentored the students on graduate school, and taught various data analysis techniques

COMMUNITY INVOLVEMENT/OUTREACH

Outreach Presentations

2025 **Williams, C. M.*,** Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in the East Portland Special Fishery Conservation Area, Jamaica. Earth Science Club of Northern Illinois

2025 **Williams, C. M.*,** Combining Environmental Monitoring and the Fossil Record to Support Coral Reef Management in the East Portland Special Fishery Conservation Area, Jamaica. Burpee Museum Fossil Fest

2023 **Williams, C. M.*,** Can fossils save the world? Combining paleontology with conservation to support a reef sanctuary in east Portland, Jamaica. Paleontological Society of Austin

Outreach Events

2023-2024 University of Texas Art Your Science Exhibit
Organized, coordinated, and set up community outreach event to display science art, and art done by scientists in the UT Jackson School community

Outreach Workshops

2021 One Love One Portland created and run by Dr. Estefania Salgado
Assisted with running workshop to involve community members and other stakeholders in east Portland Jamaica with ongoing conservation efforts at the East Portland Special Fishery Conservation Area

SKILLS

Fieldwork Skills

- PADI Open Water Diver certified, PADI Advanced Open Water Diver certified

- Reef monitoring techniques: Global Coral Reef Monitoring Network and Atlantic and Gulf Rapid Reef Assessment
- Trained in canopy research and rope climbing techniques

Programming Skills and Languages

- Software: Upper intermediate proficiency in Python and R, intermediate proficiency in QGIS
- Language: Intermediate level Spanish

WORKSHOPS

2025 Conservation Paleobiology Network (CPN) Meeting in Gainesville
Collaborated with CPN members to create nonprofit, decide bylaws, code of ethics, and governance structure during the transition of the network to an independent society

2023 Gulf of Mexico Coral Reef Symposium
Designated research priorities for Flower Garden Banks National Marine Sanctuary

2022 Unlearning Racism in Geoscience (URGE)

2021 Catalyzing Advocacy in Science and Engineering, AAAS

2021 Advanced Ecological Niche Modeling Using R, PR statistics

2021 Bermuda Institute of Ocean Science Coral Reef Ecology Course

PROFESSIONAL MEMBERSHIP

- Geological Society of America
- Paleontological Society of America
- International Coral Reef Society
- International Fossil Coral and Reef Society
- Association of Women in Geoscience
- Society for the Study of Evolution